

SolaraX

We tell you which solar panels to check first.



24/7

We watch the panels every day



Targeted

Same inspection budget, better aimed



2

Ways of finding faults, combined



Three things to remember



Checks are too far apart

Drones check once a year.
Faults grow in days.



The data is already there

Solar farms already record
how much power each row makes.

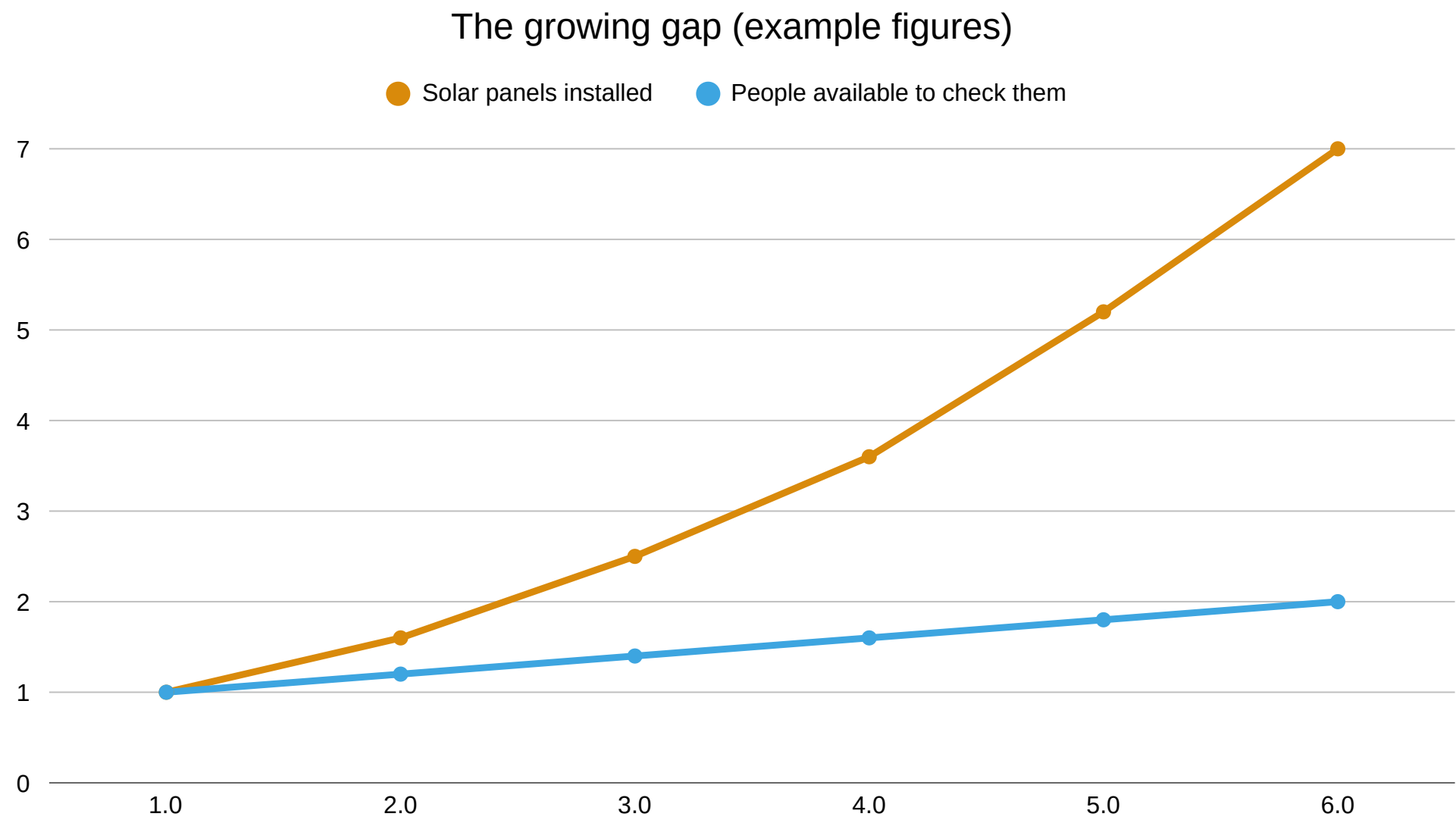


Two ways to find faults

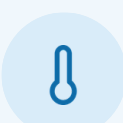
Power data finds weak panels.
Cameras find broken ones.

The drone still flies. We decide where it goes first.

Solar is growing fast. Checking it is not.



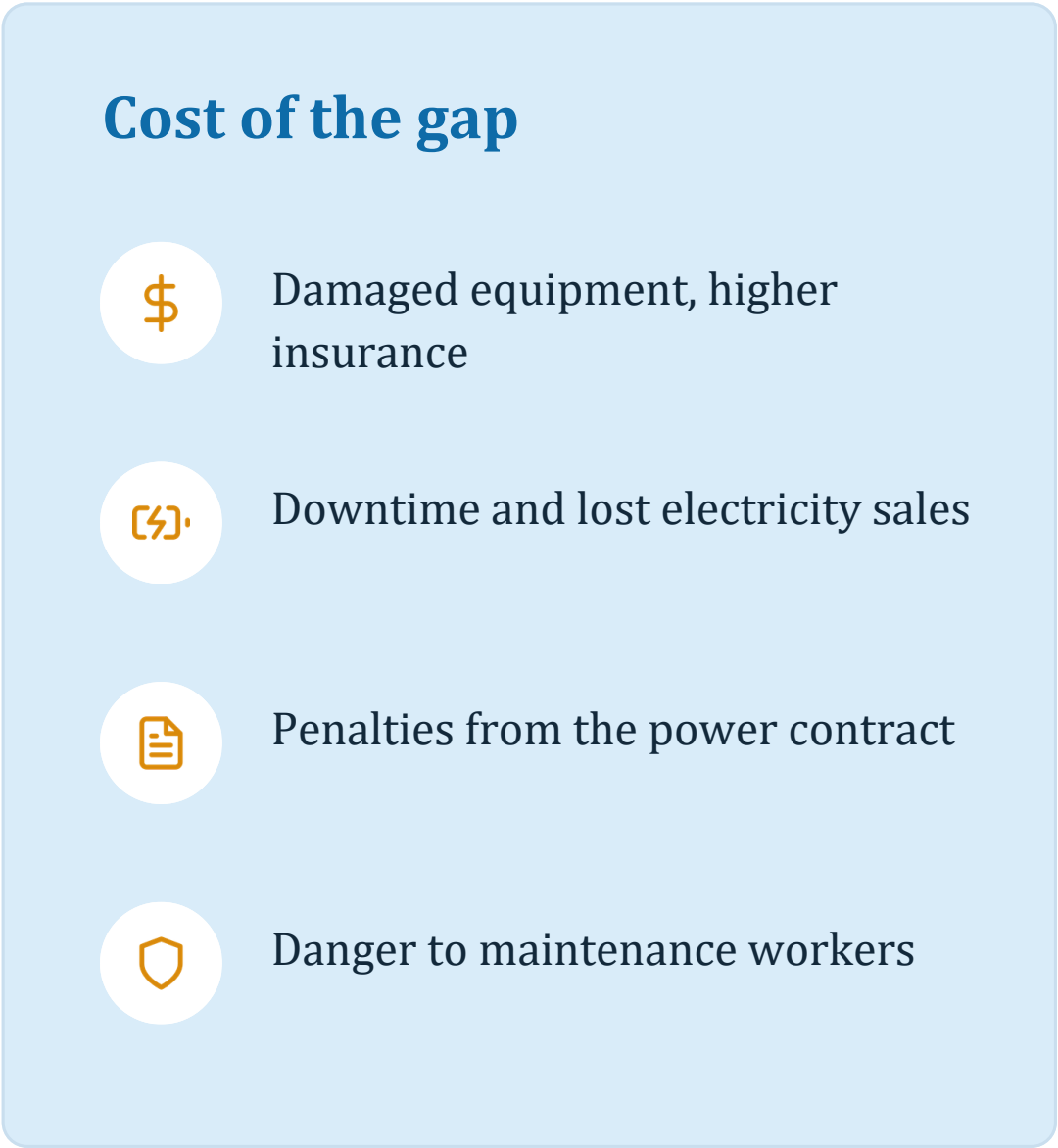
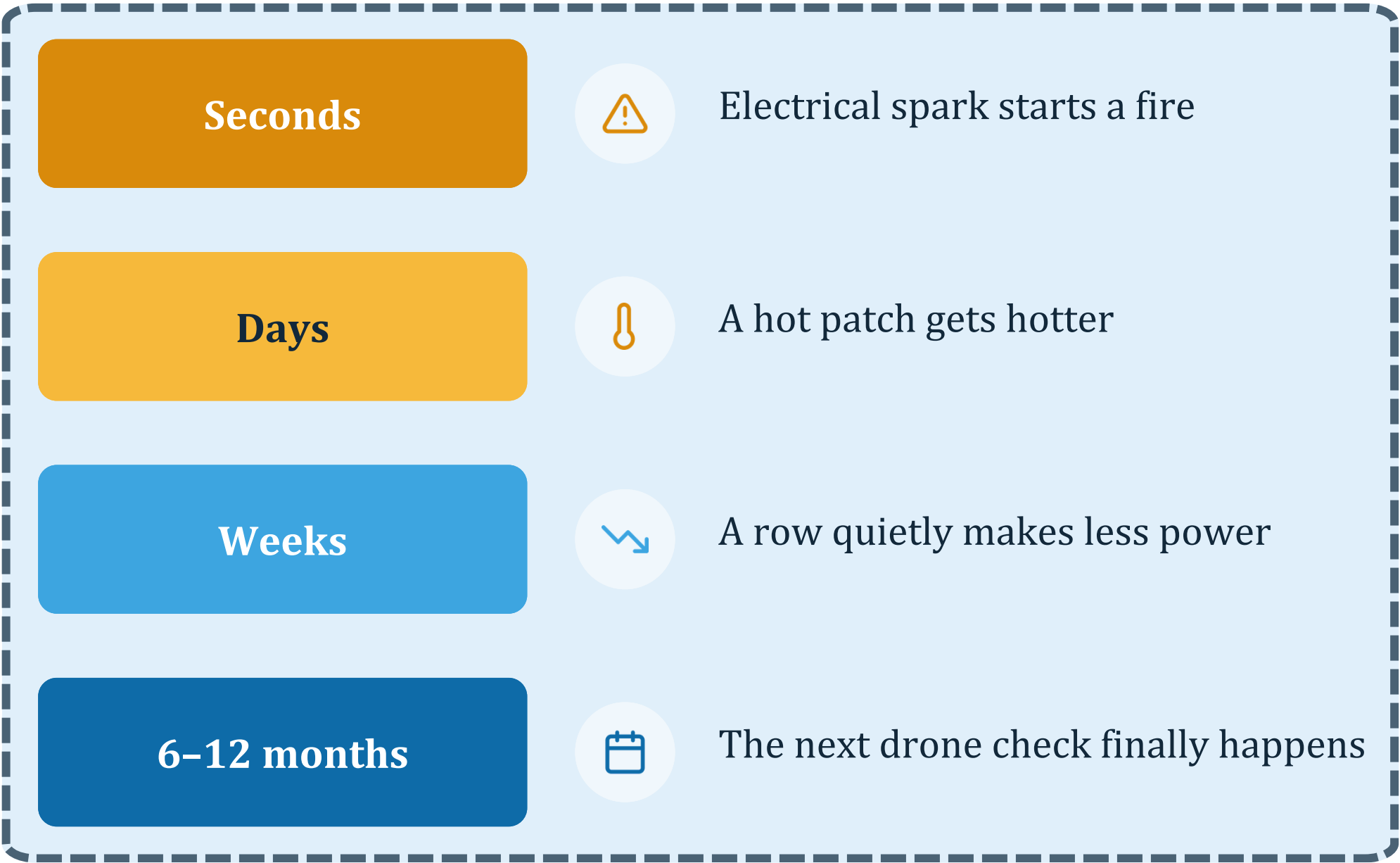
What's driving it

-  Malaysia target: 70% renewable-energy installed capacity by 2050 — NETR / MIDA
-  2025 industry signal: ACWA Power + MIDA announced plans to explore up to 12.5 GW of renewable capacity by 2040 — Reuters
-  More deployed assets mean more panels, connectors and strings to inspect
-  Inspection capacity still scales with people, flights and site visits

The growth curve is digital; inspection is still periodic. SolaraX turns existing performance data into a daily triage signal.

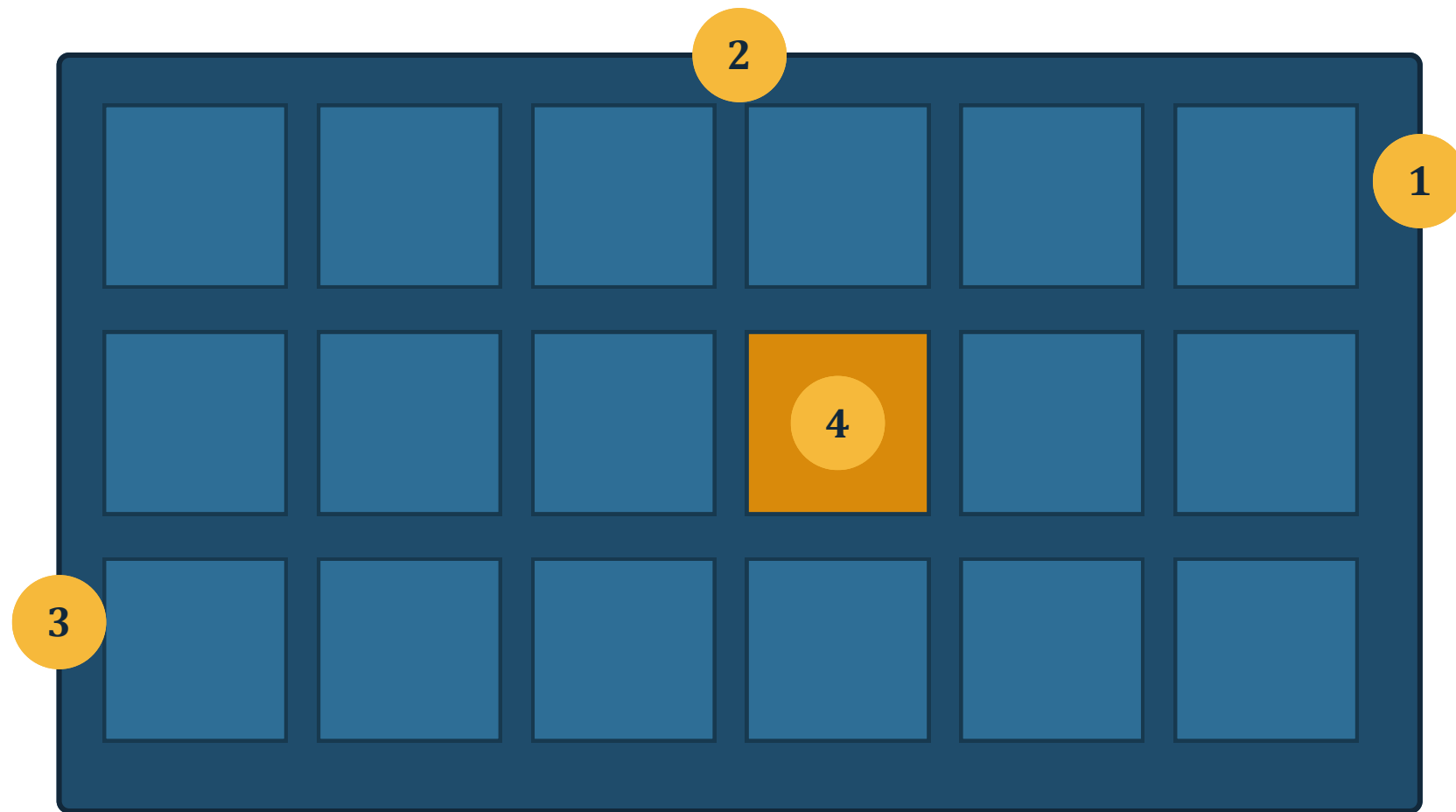
THE PROBLEM

Faults evolve daily. Inspection is periodic.



Problem statement: asset health changes every day, but thermal/RGB inspections are periodic. Faults can remain unseen between surveys, increasing downtime, safety exposure and maintenance cost.

Fires start as electrical faults, not flames



A row of solar panels

- 1 Cable connectors**
Rust and loose fittings. Most common cause.
- 2 The panel's wiring box**
Loose joints inside heat up.
- 3 Cables**
Chewed by rats, worn by sun and rubbing.
- 4 Hot patches on the panel**
Cracks and shade make one spot overheat.

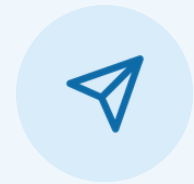
Different faults leave different signals. SolaraX combines electrical-performance anomalies with thermal/RGB evidence; it prioritises inspection, not replaces certified safety checks.

How solar farms are checked today



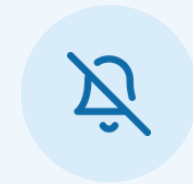
Walking the site

- Worker with a heat camera
- Impossible on a big farm
- Cost grows with farm size



Drone with a heat camera

- Works well, but only on flight day
- Cancelled by wind and rain
- Only once or twice a year



Basic control-room alarms

- Only alerts after something breaks
- Doesn't say which panel
- Doesn't say why



The opening

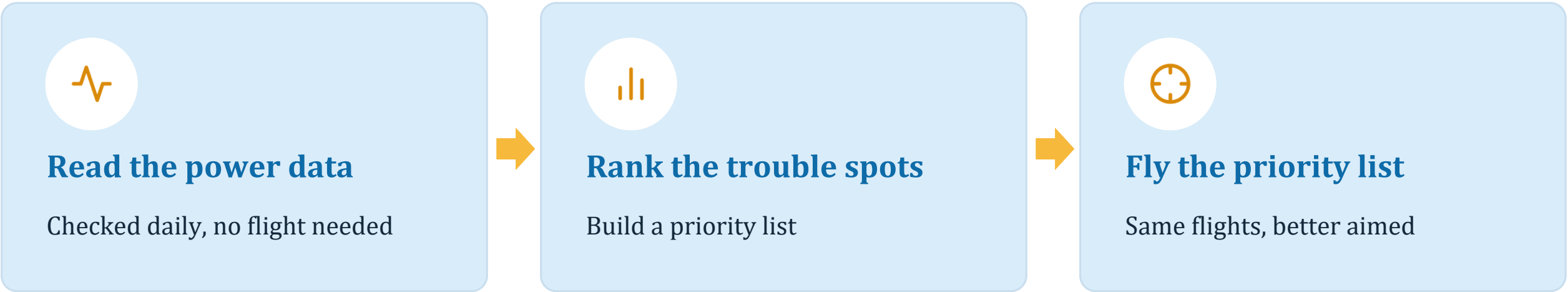
Everyone treats checking as a yearly event. Nobody watches in between.

Everyone acts after the flight. We act before.

How everyone does it now



How Solarax does it



Who does this today

- Sitemark
- Scopito
- Raptor Maps
- TÜV Rheinland MY
- KAL Energy
- DroneEye / EDX
- ALJ Global

We send drone companies work, not compete with them.

They prioritise anomalies after the flight. We prioritise the flight itself.

Where Solarax is different — without replacing drone providers

Our wedge: use daily performance data to decide what should be inspected next, then partner with existing drone operators for evidence capture.

CAPABILITY	SOLARAX	GLOBAL INSPECTION ANALYTICS	LOCAL DRONE SERVICES
Daily performance-data triage	core	Partial	usually not core
Pre-flight risk prioritisation	core	Varies by platform	Typically service / flight-plan driven
Thermal + RGB analysis	Via partner flights	core	Delivered during inspection
Verified-fix feedback loop	Designed in	Varies	Service/report dependent
Malaysia-specific deployment	Local-first	Global deployment	Strong local presence
Owns / flies drones	No — asset-light	Usually software-led	Yes

What winning looks like



≤ 24 h

to spot a problem



$\geq 85\%$

of alerts are real faults



$\geq 60\%$

less area flown



3 sites

paying customers in year 1

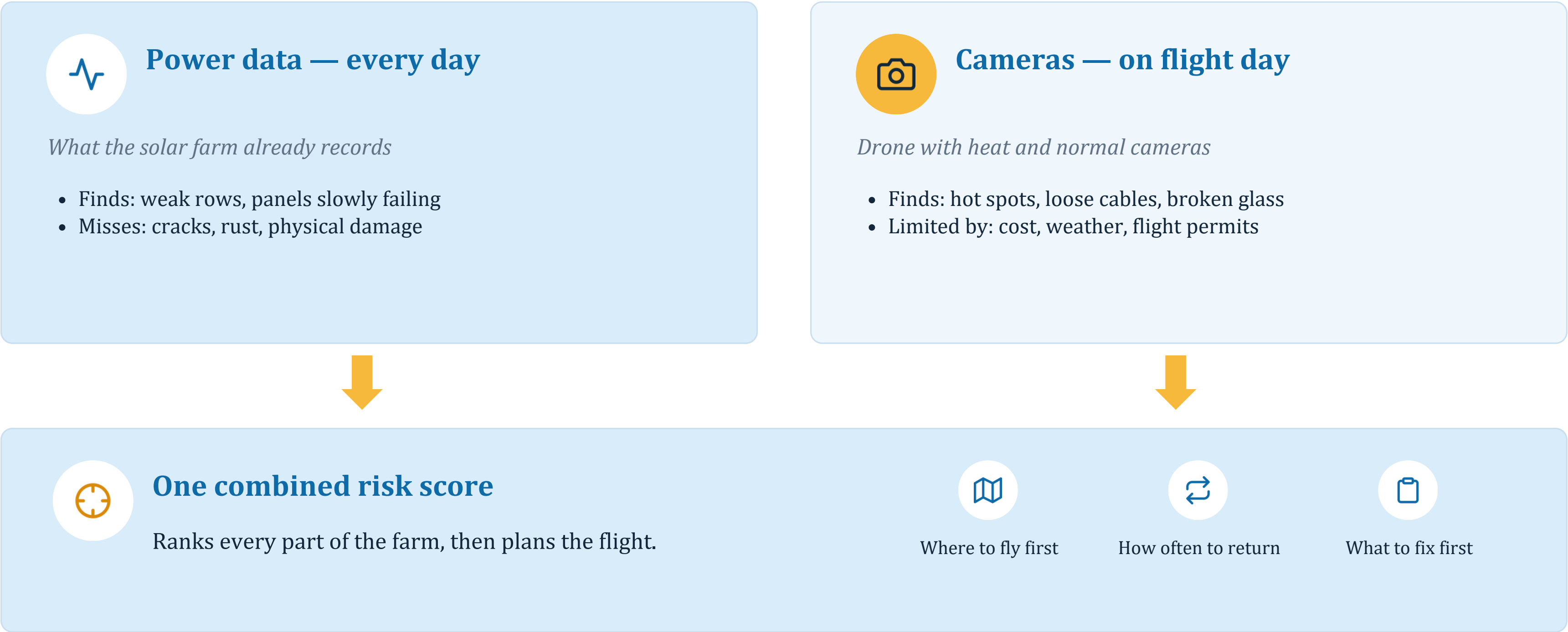
Goals

- Spot early warning signs in hours
- Send the drone to the riskiest spots first
- Give the maintenance team a ranked to-do list
- Build a Malaysian fault dataset — AI misses 20% without local data

Non-goals

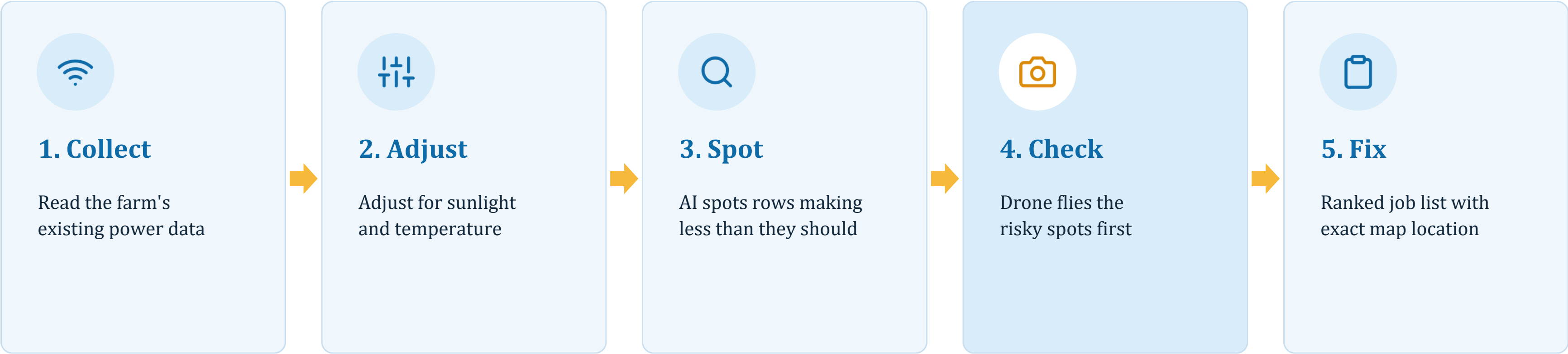
- We don't own drones
- We don't cancel the drone check
- We don't replace built-in safety switches

Two ways of looking, one priority list



The drone still flies. The data decides where and how often.

Five steps, start to finish



**Nothing to buy**

No sensors, no installation. Just permission to read the data.

**Nothing to own**

Drone companies fly. We tell them where.

How each fault is found

What we look for	What we look at	How the AI finds it	How often
Row making less power	Output vs what it should be	Pattern-learning model	Daily
One panel out of step	Panel-level power readings	Odd-one-out detection	Daily
Slow decline over time	Weeks of gradual change	Trend forecasting model	Weekly
Hot spot	Drone heat photo	Image AI, compared to neighbours	When flagged
Visible damage	Drone normal photo	Image AI finds cracks and marks	When flagged



Nothing experimental

All of this is proven, everyday technology.



Works from day one

No waiting to collect training data.



Follows the standards

International solar inspection standards.

WHY BOTH ARE NEEDED

What each method can actually find

FAULT TYPE	POWER DATA	DRONE
A row making less power than it should		
A panel slowly getting weaker		
Hot spots and failed internal parts		
Overheating cables and wiring boxes		
Cracked glass and peeling layers		
Burn marks and rust		
Loose frames, tilted panels, debris		
Dirt, weeds and pooling water		
Finds it Sometimes finds it Cannot find it		



Drones are still needed

Five of these eight faults are invisible to power data.



But flights are limited

The data decides which areas get flown first, and how often.

Illustrative unit economics — validate in pilot



RM 2,400
per megawatt, per year



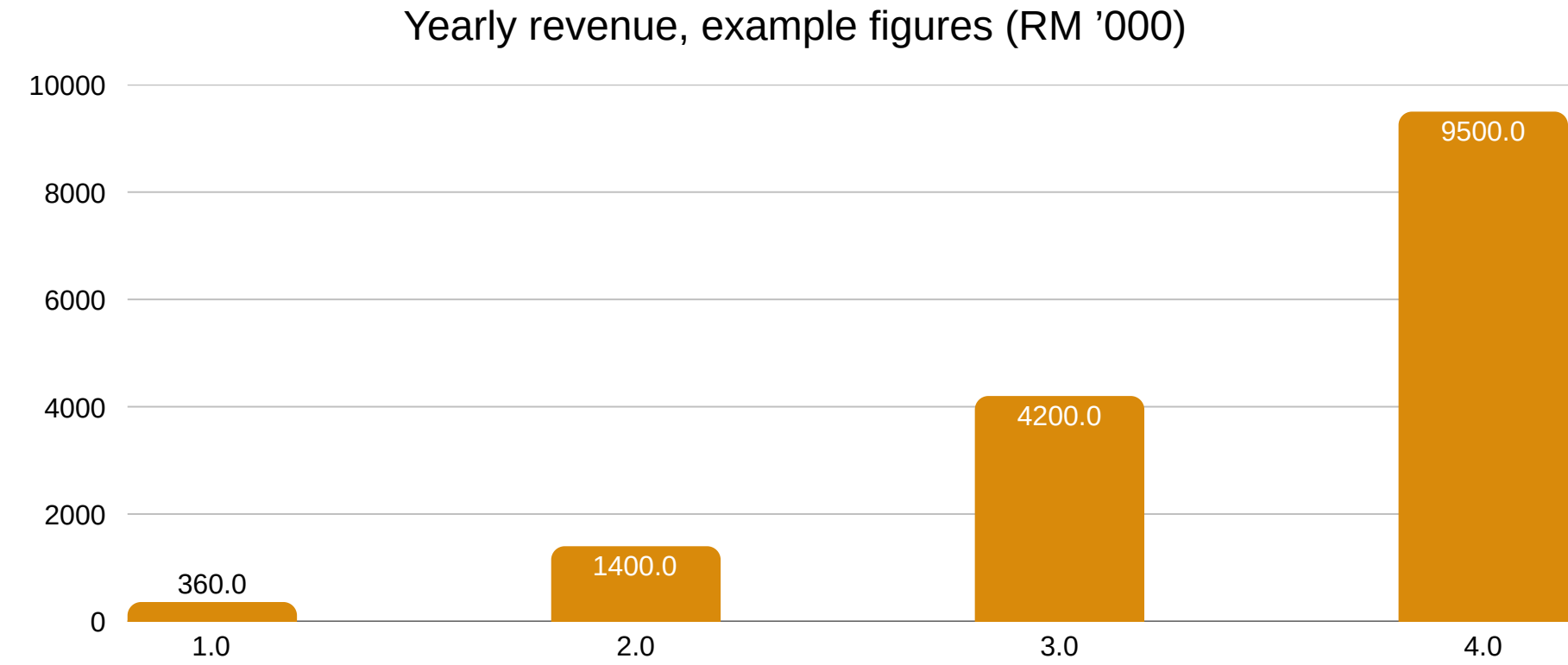
RM 120k
a year from one mid-size farm



< 12%
of that spent on running costs



~3 mo
to pay for itself



Where the money comes from

-  Subscription per megawatt watched
-  A cut of the drone flights we book
-  Reports for insurers and auditors
-  Bulk plan for owners of many farms

Start with big farms, then grow



Why this grows easily

-  A new customer needs no site visit
-  The same equipment is used region-wide
-  Every customer makes the AI smarter

12 months, four stages



Q1

Foundation

- Connect to the power data
- Build the expected-output model
- Sign a first test site

Owner: Backend Lead



Q2

Working detection

- Train the fault-spotting AI
- Dashboard and job list
- Find a first real fault

Owner: AI/ML Lead



Q3

Close the loop

- Partner with a drone company
- Train the image AI
- Confirmed faults retrain the AI

Owner: Operations Lead



Q4

Commercial

- 3 paying customers
- Reporting for insurers
- Raise funding

Owner: Commercial Lead



Rule: we don't scale until a real drone check confirms one of our alerts.

What could go wrong

	Equipment maker blocks the data	High	Read the data directly on site, with the owner's permission
	Too many false alarms	High	Rank alerts by seriousness; confirm before sending anyone
	Not enough example faults to learn from	Med	Physics model first, then a Malaysian fault dataset from confirmed finds
	A bigger company copies us	Med	Move fast locally; sell to them instead of fighting them
	Flight delayed by weather or permits	Low	The daily data checks keep running anyway
	Power data can't see physical damage	High	The yearly drone check always stays

What we aim to prove in 90 days



$\leq 24 \text{ h}$

target: flag a high-risk anomaly



$\geq 60\%$

target: less area flown



≥ 1

drone-confirmed fault



$\geq 85\%$

target: alert precision



For the solar farm

- Earlier warning between inspections
- More focused drone flights
- Ranked maintenance priorities
- Evidence trail for operators / insurers



For SolaraX

- Validate data access + detection accuracy
- Prove pre-flight prioritisation
- Measure false positives and flight reduction
- Convert pilot into first paying deployment

Good for Malaysia's clean energy push



Environmental

Longer asset life, higher yield.
Less premature module waste.
Supports NETR's RE pathway.



Social

Fewer risky rooftop climbs for workers. Protects the factories and schools with solar on the roof.



Governance

Clear inspection records build trust with banks and insurers, making solar cheaper to fund.



National relevance

Built in Malaysia, on Malaysian data, keeping the skills and jobs here.

NEXT STEPS

What we need to move



Access

One solar farm, read-only data, 90 days.



Partnership

An introduction to a Malaysian drone company.



Support

Funding for servers and development.

Decision required

Approve a 90-day paid trial on one farm. We succeed if a drone confirms a fault we flagged first.

Assumptions, targets and references



Key assumptions

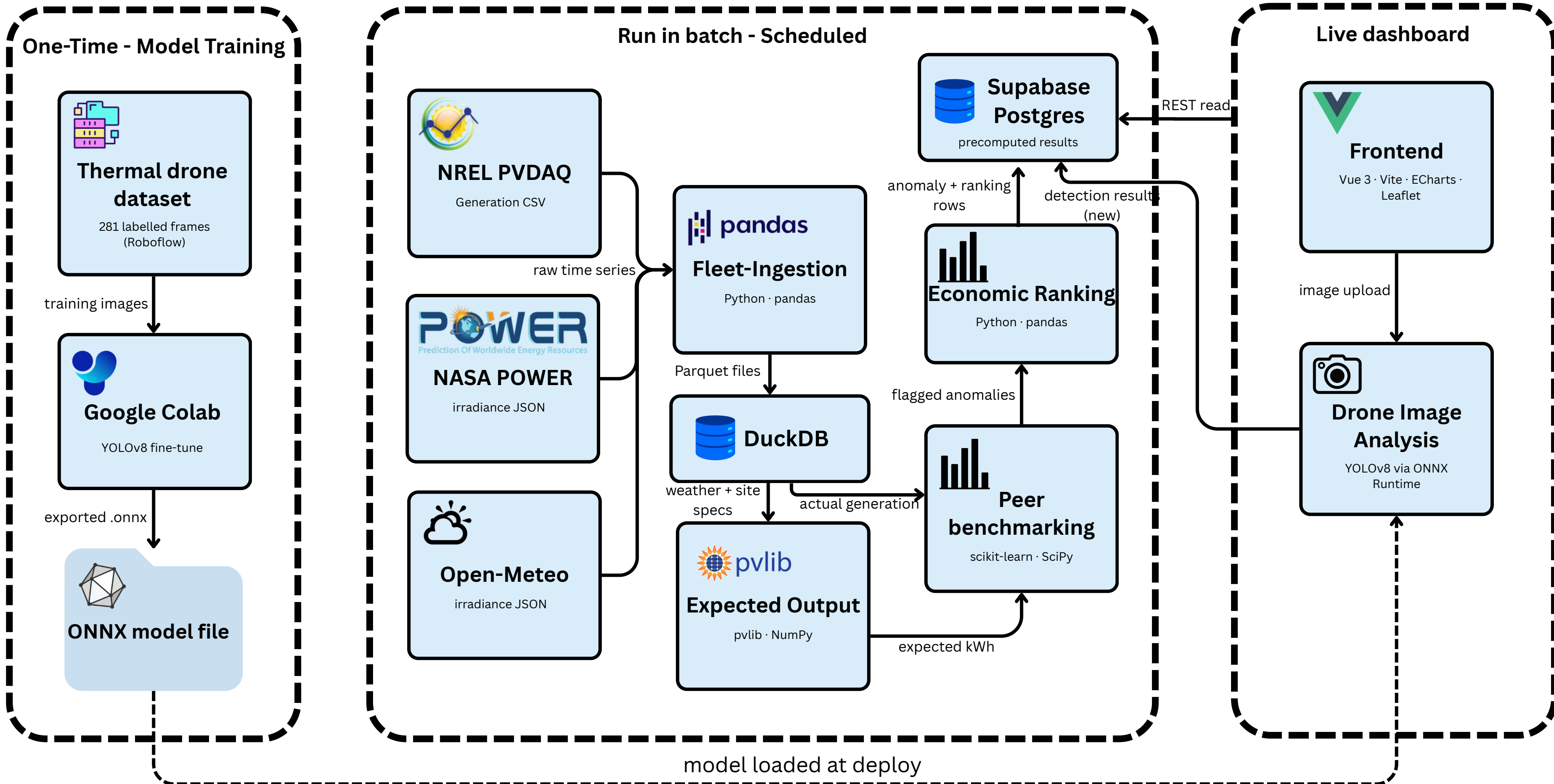
- PILOT TARGETS — NOT ACHIEVED RESULTS
- ≤ 24 h anomaly flagging
- $\geq 85\%$ alert precision
- $\geq 60\%$ less area flown
- ≥ 1 drone-confirmed fault
-
- COMMERCIAL ASSUMPTIONS
- Pricing and unit economics are illustrative until validated.
- Requires read-only access to site performance data.
- Licensed partners still perform drone / thermal inspection.
- SolaraX prioritises inspection; it does not replace protection systems.



References

- MIDA — National Energy Transition Roadmap (NETR): Malaysia targets 70% renewable-energy installed capacity by 2050.
- Reuters, 2025 — ACWA Power / MIDA plans to explore up to 12.5 GW renewable capacity in Malaysia by 2040.
- Suruhanjaya Tenaga / SEDA Malaysia — official renewable-energy and solar policy / capacity data.
- IEC 62446-3 — outdoor infrared thermography for PV systems.
- IEC 61730 — PV module safety qualification.
- MS IEC 62548 — PV array design requirements.
- Competitor sources: Sitemark; Scopito.
- Local service references: TÜV Rheinland Malaysia; KAL Energy; DroneEye / EDX; ALJ Global.
- Any other percentages or commercial figures in this deck should be treated as pilot targets / assumptions unless explicitly sourced.

System Architecture



Technology Stack



Frontend



Vue 3



Vite



ECharts



Leaflet

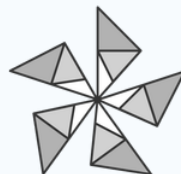
Backend/API



FastAPI



Pydantic



ONNX
Runtime

Computer Vision



YOLOv8

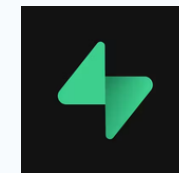


Google
Colab



Roboflow
Dataset

Data & Storage



Supabase



DuckDB



Parquet

Analytics & ML



NumPy



Python



pandas



pvlib



scikit-learn

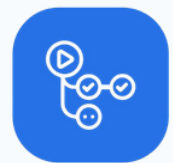


SciPy

Infra



Vercel



GitHub
Actions



pytest



Hugging
Face
Spaces

